

Life Cycle Assessment: A One-Page Framework

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What an LCA answers

A life cycle assessment (LCA) counts every greenhouse gas emitted and removed across a product's whole life. Its result is one number per unit of product. For Trailblazer, that number decides three things: a shipper's 45Z credit, its clean fuel credits, and how many removal credits each captured tonne yields.

Tallgrass owns the transport and storage stage. Its compression, pipeline and injection emissions are deducted from every shipper's result, so Tallgrass's operations change the credit value of every tonne it moves.

The four phases (ISO 14040/44)

Every credible LCA, and every rulebook Trailblazer sells into, follows the same four steps. Microsoft's 2026 removal criteria name ISO 14040/44 explicitly.

Phase	Question it answers	Trailblazer example
1. Goal and scope	What are we measuring, per what unit, and where are the boundaries?	Net CO2 removed per tonne injected, from corn field to Wyoming storage
2. Inventory (LCI)	What goes in and out at each stage?	Corn farming inputs, plant energy, 2.9 kg fermentation CO2 per gallon, booster compression power, injection metering
3. Impact assessment (LCIA)	What do those flows add up to in CO2-equivalent?	Fermentation CO2 stored, minus farming, plant, pipeline and storage emissions, minus land-use charges

Phase	Question it answers	Trailblazer example
4. Interpretation	How sure are we, and what drives the answer?	Sensitivity to land-use treatment, grid power for compression, leakage and monitoring assumptions

The phases loop. Interpretation often sends the analyst back to tighten the scope or gather better data.

One tonne, four rulebooks

The same Trailblazer tonne gets a different LCA answer under each program, because each one makes different scope choices. Knowing those choices is what makes the credit value predictable.

Program	Model	Functional unit	Land-use change for corn	What CCS earns
45Z	45ZCF-GREET (IRS Notice 2026-53)	gCO ₂ e per MJ of fuel	Excluded for fuel made after 2025 (proposed regs, Feb 2026)	About 30 CI points, taking ethanol to a CI of about 12–15
California LCFS	CA-GREET, plus CARB's CCS Protocol	gCO ₂ e per MJ of fuel	Included; CARB adds an indirect land-use value (about 20 g/MJ for corn ethanol, approximate)	The CI cut, if the plant has a certified CCS pathway
45Q	No LCA for geologic storage; Subpart RR mass balance	Tonne of CO ₂ stored	Not counted	\$85 per metered tonne
Puro.earth / Microsoft	Cradle-to-grave LCA (ISO 14040/44)	Net tonne removed	Puro charges 0.012 kg CO ₂ e per MJ of corn, about 1.5 kg per gallon	Net credits of roughly 40–50% of captured tonnes (our estimate)

The five choices that move the answer:

1. **Functional unit.** Per MJ of fuel rewards a lower-carbon gallon. Per tonne stored rewards volume. Per net tonne removed rewards the whole chain.
2. **System boundary.** Cradle-to-gate stops at the plant. Cradle-to-grave includes Tallgrass's pipeline, injection and long-term monitoring.
3. **Allocation.** A plant makes ethanol, distillers grains and corn oil. How emissions are split among them changes the fuel's CI.
4. **Biogenic carbon and land use.** Fermentation CO₂ counts as removal only if growing the corn did not cause equal emissions somewhere else. This is where 45Z, LCFS and Puro disagree most.
5. **Attributional or consequential.** Attributional LCA divides up today's emissions. Consequential LCA asks what changes because the project exists. Removal buyers are moving toward the consequential view.

Where LCAs fail, and how to defend one

Buyers and regulators reject LCAs for a handful of predictable reasons. Each one has a defense Tallgrass can build into its data.

Failure mode	Defense
Default values instead of measured data	Use metered injection volumes, actual compression power and site monitoring data
Double counting the same tonne	Keep one attribute ledger, so no tonne carries both a clean fuel credit and a removal credit
Hidden land-use effects	State the land-use treatment openly and show results with and without it
False precision	Report a range and the uncertainty deduction, as Puro and Microsoft require
Ignoring reversals	Tie results to the Class VI permit, monitoring plan and a buffer or replacement commitment

The working rule: an LCA is only as defensible as its weakest stage, and Tallgrass controls the stage buyers can audit most easily.

Sources: Microsoft, *Criteria for High-Quality Carbon Dioxide Removal, 2026 Edition*; Puro.earth, *Geologically Stored Carbon v.5*; [26 USC 45Z](#); [CARB CCS Protocol](#); [CarbonPlan](#). Facts are carried over from the Trailblazer Low-Carbon Market Strategy report; the CARB land-use value is approximate.